

## Analysis of Singapore's Public Housing (HDB) Market (2010–2022)

### Data Sources

- Data from Kaggle: *Resale HDB Flat Prices 2000–2022*
- The dataset was divided into 4 tables, then cleaned and merged into 2 main tables:
  - **Table 1:** Covers data from 2010 to April 2022, with the *Remaining Lease* column removed
  - **Table 2:** Focuses on *Remaining Lease* (remaining lease duration), only available from 2015 onwards

### Project Objectives

- Analyze resale trends of Singapore public housing (HDB flats) from 2010 to early 2022
- Identify active market segments, flat types, and high-transaction regions
- Evaluate the impact of factors such as COVID-19, remaining lease, and government policies on market behavior
- Present analytical findings to support better decision-making for homebuyers, investors, and agents

### Approach & Analytical Steps

1. **Data Processing:** Cleaning, standardizing, and merging based on time, flat type, region, and lease
2. **Overview Analysis:**
  - Annual transaction volume and value
  - Deep-dive comparison of key years (2010, 2020, 2021)
3. **Regional/Location Analysis:**
  - Focus on active areas: Jurong West, Sengkang, Tampines, Woodlands, Yishun
  - Highlight: Punggol shows significant recent growth
4. **Flat Type Analysis:**
  - The mass-market segment (SGD 300k–650k) holds the largest share of transactions
  - Most transactions involve 3–5 room flats, especially 4-room types
  - Multi-generation flats saw a sharp decline due to mismatched demand
5. **Remaining Lease Analysis:**
  - Flats with >70 years remaining have the highest value and transaction volume

- Flats with <50 years lease see low demand due to liquidity and value concerns

### **Key Insights**

- Market downturn from 2010–2014, with strong recovery from 2015–2021
- COVID-19 didn't hinder the market; it actually boosted activity in 2020–2021
- Increasing trend of buyers preferring newer flats (built after 2000), reflecting higher living standards
- Transactions concentrated in mid-tier flats ( $\leq 12$  storeys) in the mass market segment
- *Remaining Lease* became a critical evaluation factor post-2015

### **Recommendations & Applications**

#### **For Owner-Occupiers:**

- Prioritize mass-market flats with 3–5 rooms, built after 2000
- Choose flats with >70 years remaining lease for long-term value
- Favor fast-developing areas like Sengkang, Tampines, and Punggol
- For budget-conscious buyers, opt for flats on the 12th floor or lower

#### **For Investors:**

- Invest in regions with high infrastructure/population growth (e.g., Punggol)
- Avoid flats with <50 years lease due to depreciation and resale difficulty
- Monitor HDB policies closely for flexible investment strategies
- Focus on the mass-market segment – broad target market and high liquidity

### **Project Deliverables:**

- **Power BI (.pbix)** file: Interactive visualization by time, location, flat type, and lease
- **Detailed Report (.docx)**: Quantitative analysis and deep insights
- **Presentation (.pptx)**: Summary for quick sharing and presentation purposes